

Question 1

Import pandas under the alias `pd`.

Solution

```
import pandas as pd
```

Explanation

- `import` loads the library.
 - `as pd` creates an alias.
 - Nearly every Pandas program starts with this line.
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Question 2

Print the version of pandas.

Solution

```
import pandas as pd

print(pd.__version__)
```

Sample Output

```
2.3.0
```

(Version depends on your installation.)

Explanation

`__version__` returns the installed Pandas version.

Question 3

Print all version information of libraries required by Pandas.

Solution

```
import pandas as pd

pd.show_versions()
```

Explanation

Shows:

- Python version
- Pandas version
- NumPy version
- Matplotlib version
- OS information

- Installed optional libraries

Useful for debugging.

Question 4

Create the DataFrame.

Given:

```
import numpy as np

data = {
    'animal': ['cat', 'cat', 'snake', 'dog', 'dog',
               'cat', 'snake', 'cat', 'dog', 'dog'],
    'age': [2.5, 3, 0.5, np.nan, 5,
            2, 4.5, np.nan, 7, 3],
    'visits': [1, 3, 2, 3, 2,
              3, 1, 1, 2, 1],
    'priority': ['yes', 'yes', 'no', 'yes', 'no',
                 'no', 'no', 'yes', 'no', 'no']
}

labels=['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

Solution

```
df = pd.DataFrame(data, index=labels)

print(df)
```

Output

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no
d	dog	NaN	3	yes
e	dog	5.0	2	no
f	cat	2.0	3	no
g	snake	4.5	1	no
h	cat	NaN	1	yes
i	dog	7.0	2	no
j	dog	3.0	1	no

Explanation

index=labels assigns custom row labels.

Question 5

Display summary information.

Solution

```
df.info()
```

Sample Output

```
<class 'pandas.core.frame.DataFrame'>
Index: 10 entries

Columns:
animal
age
visits
priority

Non-Null Counts
Data types
Memory usage
```

Explanation

`info()` provides:

- rows
- columns
- missing values
- data types
- memory usage

Question 6

Return the first three rows.

Solution

```
df.head(3)
```

Output

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no

Explanation

`head(3)` displays the first three rows.

Question 7

Select only the `animal` and `age` columns.

Solution

```
df[['animal', 'age']]
```

Output

	animal	age
a	cat	2.5
b	cat	3.0
c	snake	0.5
...	...	...

Explanation

Selecting multiple columns requires a **list of column names**.

Question 8

Select rows `[3, 4, 8]` and columns `animal` and `age`.

Solution

```
df.loc[df.index[[3, 4, 8]], ['animal', 'age']]
```

Alternative

```
df.iloc[[3, 4, 8]][['animal', 'age']]
```

Output

	animal	age
d	dog	NaN
e	dog	5.0
i	dog	7.0

Explanation

- `loc` uses labels.
 - `iloc` uses integer positions.
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Question 9

Select rows where visits > 3.

Solution

```
df[df['visits'] > 3]
```

Output

```
Empty DataFrame
```

Explanation

In the given dataset, the maximum number of visits is 3, so no row satisfies `visits > 3`.

Question 10

Select rows where age is missing (NaN).

Solution

```
df[df['age'].isna()]
```

Alternative

```
df[df['age'].isnull()]
```

Output

	animal	age	visits	priority
d	dog	NaN	3	yes
h	cat	NaN	1	yes

Explanation

- `isna()` returns `True` for missing values.
- Filtering with it returns only rows containing missing ages.